

# MOTION IN A STRAIGHT LINE

## Important formulas and concepts directly from the NCERT-

- Study of motions along a straight line is known as '**Rectilinear motion**'.
- In **kinematics**, we study ways to describe motion without going into the causes of motion.
- '**Frame of reference**' is a place from where we take observation.

### ❖ Position, path length and displacement-

- Location of object from frame of reference is k/as its 'position'.
- Motion is change in position of an object with time.
- Rest and motion are relative terms, a body can be in rest or motion with respect to frame of reference.
- **Distance** is the total path length covered by an object and **displacement** is the shortest distance b/w any two points.
- Difference b/w distance and displacement-

| Distance                                                                                                                                                                                                                                             | Displacement                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>1) Property of moving object.</li><li>2) Always positive, can't be negative.</li><li>3) Always added.</li><li>4) It can't increase with time.</li><li>5) Scalar quantity.</li><li>6) Only magnitude.</li></ul> | <ul style="list-style-type: none"><li>1) Shift in position.</li><li>2) Can be positive, negative or zero.</li><li>3) It is net shift in position (not always added)</li><li>4) It can increase with time.</li><li>5) Vector quantity(direction from initial to final position)</li><li>6) Both magnitude and direction.</li></ul> |

- **The magnitude of displacement may or may not be equal to the path length traversed by an object.**
- The magnitude of displacement may be zero but distance can't be zero.

### ❖ Average speed and average velocity-

- Average velocity is the change in position or displacement divided by the time.

$$v = \Delta x / \Delta t$$

- **SI unit of velocity is m/s. (km/h also used in many applications)**

- Average speed is the total path length travelled divided by the total time.

$$v = \text{total path length} / \text{total time interval}$$

- average speed doesn't tell us in which direction an object is moving.
- **Average speed is also always positive.**
- **Average velocity can be positive or negative.**
- **If the motion of an object is along a straight line and in the same direction, the magnitude of displacement is equal to the total path length. And magnitude of average velocity is equal to the average speed.**
- Average speed is greater than or equal to the average velocity.
- **Speed is in general , greater than the magnitude of velocity**(because path length is greater than magnitude of displacement)

### ❖ Instantaneous velocity and speed-

- Velocity at any instant of time is known as instantaneous velocity.  
It is rate of change in position.

$$v = dx/dt$$

- **For uniform motion, velocity is the same as the average velocity at all instants.**
- Instantaneous speed or simply speed is equal to the magnitude of velocity.
- Velocity is the slope of tangent to the displacement-time curve/graph.

### ❖ Acceleration-

- Rate of change of velocity with time.

➤ **Average acceleration =  $\Delta v / \Delta t$**

➤ **SI unit =  $m/s^2$**

➤ **Instantaneous acceleration =  $dv/dt$**

➤ **Acceleration is the slope of tangent to the velocity-time curve.**

➤ **Area under velocity-time curve represents the displacement and area under acceleration-time curve represents the velocity.**

❖ **Kinematic equations for uniformly accelerated motion-**

1)  $v = v_0 + at$

2)  $x = v_0 t + \frac{1}{2} at^2$

3)  $v^2 = v_0^2 + 2ax$

❖ **relative velocity-**

➤ **velocity of object A relative to object B,**

$$v_{AB} = v_A - v_B$$

➤ **if two bodies are in same direction then, for **relative velocity** , **individual velocities are subtracted.****

**if two bodies are in opposite direction then, for **relative velocity** , **individual velocities are added.****

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## SUMMARY

1. An object is said to be in *motion* if its position changes with time. The position of the object can be specified with reference to a conveniently chosen origin. For motion in a straight line, position to the right of the origin is taken as positive and to the left as negative.
2. *Path length* is defined as the total length of the path traversed by an object.
3. *Displacement* is the change in position :  $\Delta x = x_2 - x_1$ . Path length is greater or equal to the magnitude of the displacement between the same points.
4. An object is said to be in *uniform motion* in a straight line if its displacement is equal in equal intervals of time. Otherwise, the motion is said to be *non-uniform*.
5. *Average velocity* is the displacement divided by the time interval in which the displacement occurs :

$$\bar{v} = \frac{\Delta x}{\Delta t}$$

On an  $x$ - $t$  graph, the average velocity over a time interval is the slope of the line connecting the initial and final positions corresponding to that interval.

6. *Average Speed* is the ratio of total path length traversed and the corresponding time interval.

The average speed of an object is greater or equal to the magnitude of the average velocity over a given time interval.

7. *Instantaneous velocity* or simply velocity is defined as the limit of the average velocity as the time interval  $\Delta t$  becomes infinitesimally small :

$$v = \lim_{\Delta t \rightarrow 0} \bar{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

The velocity at a particular instant is equal to the slope of the tangent drawn on position-time graph at that instant.

8. *Average acceleration* is the change in velocity divided by the time interval during which the change occurs :

$$\bar{a} = \frac{\Delta v}{\Delta t}$$

9. *Instantaneous acceleration* is defined as the limit of the average acceleration as the time interval  $\Delta t$  goes to zero :

$$a = \lim_{\Delta t \rightarrow 0} \bar{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$

The acceleration of an object at a particular time is the slope of the velocity-time graph at that instant of time. For uniform motion, acceleration is zero and the  $x$ - $t$  graph is a straight line inclined to the time axis and the  $v$ - $t$  graph is a straight line

parallel to the time axis. For motion with uniform acceleration,  $x-t$  graph is a parabola while the  $v-t$  graph is a straight line inclined to the time axis.

10. The area under the velocity-time curve between times  $t_1$  and  $t_2$  is equal to the displacement of the object during that interval of time.
11. For objects in uniformly accelerated rectilinear motion, the five quantities, displacement  $x$ , time taken  $t$ , initial velocity  $v_0$ , final velocity  $v$  and acceleration  $a$  are related by a set of simple equations called *kinematic equations of motion* :

$$v = v_0 + at$$

$$x = v_0 t + \frac{1}{2} at^2$$

$$v^2 = v_0^2 + 2ax$$

if the position of the object at time  $t = 0$  is 0. If the particle starts at  $x = x_0$ ,  $x$  in above equations is replaced by  $(x - x_0)$ .

### important points from NCERT summary-

- an object is said to be in *uniform motion* in a straight line if its displacement is equal in equal intervals of time. Otherwise, motion is said to be *non-uniform*.

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### POINTS TO PONDER

1. The path length traversed by an object between two points is, in general, not the same as the magnitude of displacement. The displacement depends only on the end points; the path length (as the name implies) depends on the actual path. In one dimension, the two quantities are equal only if the object does not change its direction during the course of motion. In all other cases, the path length is greater than the magnitude of displacement.
2. In view of point 1 above, the average speed of an object is greater than or equal to the magnitude of the average velocity over a given time interval. The two are equal only if the path length is equal to the magnitude of displacement.
3. The origin and the positive direction of an axis are a matter of choice. You should first specify this choice before you assign signs to quantities like displacement, velocity and acceleration.
4. If a particle is speeding up, acceleration is in the direction of velocity; if its speed is decreasing, acceleration is in the direction opposite to that of the velocity. This statement is independent of the choice of the origin and the axis.
5. The sign of acceleration does not tell us whether the particle's speed is increasing or decreasing. The sign of acceleration (as mentioned in point 3) depends on the choice of the positive direction of the axis. For example, if the vertically upward direction is chosen to be the positive direction of the axis, the acceleration due to gravity is negative. If a particle is falling under gravity, this acceleration, though negative, results in increase in speed. For a particle thrown upward, the same negative acceleration (of gravity) results in decrease in speed.
6. The zero velocity of a particle at any instant does not necessarily imply zero acceleration at that instant. A particle may be momentarily at rest and yet have non-zero acceleration. For example, a particle thrown up has zero velocity at its uppermost point but the acceleration at that instant continues to be the acceleration due to gravity.
7. In the kinematic equations of motion [Eq. (3.11)], the various quantities are algebraic, i.e. they may be positive or negative. The equations are applicable in all situations (for one dimensional motion with constant acceleration) provided the values of different quantities are substituted in the equations with proper signs.
8. The definitions of instantaneous velocity and acceleration (Eqs. (3.3) and (3.5)) are exact and are always correct while the kinematic equations (Eq. (3.11)) are true only for motion in which the magnitude and the direction of acceleration are constant during the course of motion.

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